Lesson 1: Velocity and Time

Edexcel IAL Bridge - Sample Worksheet

Physics - WPH11 (2018 specification)

Total: 20 marks **Suggested time: 30 minutes**
Answer all questions. Show working. Give units and sensible significant figures.

Original practice; not an official Pearson paper. Calculator and ruler permitted.

Part A: MULTIPLE CHOICE QUESTIONS — 4 marks

Choose one answer per question. Write your answers in the grid.

1. The gradient of a velocity-time graph gives

A. distance.	**B.** acceleration.
C. displacement.	**D.** average speed.

2. An area below the time axis on a velocity-time graph contributes

A. negative displacement.	**B.** negative distance.
C. positive acceleration.	**D.** zero speed.

3. A trolley has velocity -3.0 m s^{-1} and acceleration $+2.0$ m s^{-2}. Its speed is

A. increasing.	**B.** zero.
C. decreasing.	**D.** constant.

4. Which condition is required to use $s = ut + \frac{1}{2}at^2$ for one-dimensional motion?

A. speed is constant.	**B.** initial velocity is zero.
C. displacement is positive.	**D.** acceleration is constant.

1:	2:	3:	4:

Part B: FILL IN THE BLANKS — 4 marks

Write one word, unit or formula in each numbered answer box. No word bank is supplied.

1. Velocity is a ____[1]____ quantity because direction matters.
2. The SI unit of acceleration is ____[2]____.
3. On a velocity-time graph, signed area gives ____[3]____.
4. Average speed equals total ____[4]____ divided by total time.

1:	2:	3:	4:

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Part C: STRUCTURED QUESTIONS **12 marks**

5. From a graph to an equation **(7)**

A trolley moves along a straight track with initial velocity u and constant acceleration a. Its velocity after elapsed time t is v and its displacement is s.

a. Deriving the relationships

i. Starting with the definition of acceleration, derive $v = u + at$. **(2)**

ii. Using the area under a straight velocity-time graph, derive $s = ut + \frac{1}{2}at^2$ for positive u and positive a. You may draw your own sketch as part of your working. **(3)**

b. Using the relationship

i. The trolley has $u = 2.0$ m s^{-1} and $a = 1.5$ m s^{-2}. Calculate its displacement after 4.0 s. **(2)**

Total for Question 5 = 7 marks

班級(Class): _____ 學號(No): _____

6. Direction and experimental uncertainty **(5)**

a. A trolley reverses direction

A trolley has initial velocity -4.0 m s^{-1} and constant acceleration $+2.0$ m s^{-2} for 6.0 s. Take motion to the right as positive.

i. Calculate the time when the trolley reverses direction. **(1)**

ii. Calculate its displacement during the 6.0 s. **(1)**

iii. Calculate the total distance travelled during the 6.0 s. **(1)**

b. A motion-sensor experiment

In a separate experiment, a velocity-time graph has best-fit gradient 0.80 m s^{-2}. The steepest and shallowest acceptable gradients are 0.88 m s^{-2} and 0.72 m s^{-2}.

i. Estimate the absolute uncertainty in acceleration using half the gradient range, then calculate the percentage uncertainty. **(2)**

Total for Question 6 = 5 marks

END OF PAPER

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